

SUPERVISED MACHINE LEARNING ALGORITHM:

Linear Regression is a fundamental supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous values by fitting a straight line that best represents the data.

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression, which predicts continuous values it predicts the probability that an input belongs to a specific class.

A decision tree is a supervised learning algorithm used for both classification and regression tasks. It has a hierarchical tree structure which consists of a root node, branches, internal nodes and leaf nodes. It works like a flowchart that helps in making step-by-step decisions.

Random Forest is an ensemble learning method that combines multiple decision trees to produce more accurate and stable predictions. It can be used for both classification and regression tasks, where regression predictions are obtained by averaging the outputs of several trees.

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. It tries to find the best boundary known as hyperplane that separates different classes in the data. It is useful when you want to do binary classification like spam vs. not spam or cat vs. dog.

K-Nearest Neighbour (KNN) is a simple and widely used machine learning technique for classification and regression tasks. It works by identifying the K closest data points to a given input and making predictions based on the majority class or average value of those neighbours.

Gradient Boosting is a boosting algorithm and here each new model is trained to minimize the loss function such as mean squared error or cross-entropy of the previous model using gradient descent. In each iteration the algorithm computes the gradient of the loss function with respect to predictions and then trains a new weak model to predict this gradient. Predictions of the new model are then added to the ensemble (all models prediction) and the process is repeated until a stopping criterion is met.

UNSUPERVISED MACHINE LEARNING ALGORITHM:

K-Means Clustering groups similar data points into clusters without needing labelled data. It is used to uncover hidden patterns when the goal is to organize data based on similarity.

Hierarchical Clustering is an unsupervised learning technique that groups data into a hierarchy of clusters based on similarity. It builds a tree-like structure (dendrogram) that helps visualize relationships and decide the optimal number of clusters.

DBSCAN is a density-based clustering algorithm that groups data points that are closely packed together and marks outliers as noise based on their density in the feature space. It identifies clusters as dense regions in the data space separated by areas of lower density. Unlike K-Means or hierarchical clustering which assumes clusters are compact and spherical, DBSCAN perform well in handling real-world data irregularities.

Apriori Algorithm is a data mining technique used to identify items that frequently appear together in large datasets. It helps discover relationships and association rules between items, making it widely used in market basket analysis.

The FP-Growth (Frequent Pattern Growth) algorithm efficiently mines frequent item sets from large transactional datasets. Unlike the Apriori algorithm which suffers from high computational cost due to candidate generation and multiple database scans. FP-Growth avoids these inefficiencies by compressing the data into an FP-Tree (Frequent Pattern Tree) and extracts patterns directly from it.

PCA (Principal Component Analysis) is a dimensionality reduction technique and helps us to reduce the number of features in a dataset while keeping the most important information. It changes complex datasets by transforming correlated features into a smaller set of uncorrelated components.